Muthiah Shanmugam

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Senior CAE Engineer — Vehicle Dynamics & Controls Simulation

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PROFILE

Senior CAE Engineer with 11+ years of full-vehicle multibody simulation, ride comfort and handling analysis, and vehicle dynamics model validation across OEM passenger car and Tier-1 supplier programs. Integrates active chassis controllers into SiL/MiL co-simulation (ADAMS/Controls, MATLAB/Simulink, FMU/FMI), enabling virtual validation without physical prototype hardware. Develops Python/PyQt tools for batch simulation and KPI evaluation; part-time doctoral research at KU Leuven on semi-active suspension control.

SKILLS

Modelling & Simulation	Full-vehicle MBS (MSC ADAMS/Car, SIMPACK) · Driveline & subsystem MBS (ADAMS/View, Gear AT) · K&C, ride comfort, handling and durability analysis · IPG CarMaker · 1D system modelling (MATLAB/Simulink)
Controls & Validation	SiL/MiL co-simulation & controls integration (ADAMS/Controls, MATLAB/Simulink, FMU/FMI) · Semi-active damper control (skyhook, LQR, adaptive, MPC) · Signal architecture & closed control loops · Model-to-measurement correlation (K&C test rig, vehicle test data) · Variant & sensitivity studies · dSPACE ControlDesk
Programming & Tooling	Python 3 (PyQt) · MATLAB scripting · C/C++ (basic) · Git/GitHub · Tortoise SVN
Languages	English — C2 · German — B2 · Tamil — native

EXPERIENCE

Senior CAE Engineer — Vehicle Dynamics & Controls Simulation
Apr 2015 – present

Akkodis GmbH (formerly AKKA Technologies / MBtech Group) · Sindelfingen, Germany

Senior since 2019 · Technical lead on 1–2 concurrent projects with teams of 2–3 engineers

- Enabled virtual validation of active chassis controllers without prototype hardware by building closed-loop SiL/MiL co-simulation: compiled Simulink controller FMUs linked to ADAMS/Car vehicle models, with defined signal architecture and closed control loops, validated against reference datasets.
- Achieved series-level model accuracy across 5+ suspension variants on passenger and commercial vehicle platforms by correlating ADAMS/Car and SIMPACK models against K&C test-rig and vehicle test data: tuned kinematic hard points, bushing rates, spring and bump-stop positions, camber/toe characteristics and tyre parameters, and traced deviations to their root cause.
- Provided the basis for chassis hardware and setup decisions from concept phase to series maturity through K&C, ride comfort, handling and durability variant studies against OEM targets (load cases unladen to fully laden); presented results in OEM technical reviews.
- Cut post-processing effort from weeks to hours with a Python/PyQt simulation process manager that automates batch K&C, ride and handling runs in ADAMS/Car (hundreds of solver runs per study), including KPI evaluation, sensitivity analysis, parameter optimisation and plot/report export.
- Verified the engagement behaviour of an electrically actuated axle-disconnect system in closed-loop MiL co-simulation (ADAMS/View cam mechanism with MATLAB/Simulink engagement logic); evaluated park-lock pawl geometries for engagement, kinematics and contact, and drove differential concept optimisation from noise-critical operating states.

Working Student — Digital Product R&D, Mercedes-Benz Cars
Dec 2013 – Aug 2014

Daimler AG · Böblingen, Germany

- Contributed to a digital development environment for vehicle door structures in the early design phase; performed mesh adaptation to support rapid design changes.

Project Engineer — Engineering Design Services
Jun 2011 – Jul 2013

Wipro Technologies · Bangalore, India

- Developed subsystem and full-vehicle FE models for crash, occupant safety and pedestrian safety (HyperMesh, ANSA), including meshing and result evaluation.

DOCTORAL RESEARCH

Doctoral Researcher (part-time) — Mechanical Engineering
Jan 2023 – present · expected completion 2027

KU Leuven · Leuven, Belgium

- Quantified the effect of vehicle-mass variation on fixed versus adaptive semi-active suspension control strategies on a quarter-car model (AIP Conf. Proc., 2025).
- Investigated the P-LQR method for semi-active suspension control (AIP Conf. Proc., 2023).
- Benchmarked skyhook, LQR, adaptive and MPC controllers in IPG CarMaker co-simulation; validation on a test vehicle in preparation.

EDUCATION

M.Eng. Automotive Systems — Hochschule Esslingen, Germany
2013 – 2015 · Grade 1.8

Focus: multibody simulation, vehicle handling, system modelling and simulation, control engineering

Master's thesis, MBtech Group GmbH & Co. KGaA, MBS Department (Sep 2014 – Feb 2015): *"Simulation-based development and adaptation of control strategy for semi-active damper on full vehicle model"*. Developed a semi-active damper control strategy on a full-vehicle model via MATLAB/Simulink – ADAMS/Car co-simulation.

B.Eng. Mechatronics — Anna University of Technology, India
2008 – 2011 · CGPA 9.62/10 · University Gold Medal

PUBLICATIONS

- Shanmugam, M., Demeester, E., Tjahjowidodo, T. (2025). Comparative study of fixed and adaptive control strategy in semi-active suspension systems under influence of vehicle mass variation using a quarter car model. *AIP Conference Proceedings* 3262(1), 030008. DOI: 10.1063/5.0246102
- Shanmugam, M., Demeester, E., Tjahjowidodo, T. (2023). Semi-active suspension control using P-LQR method. *AIP Conference Proceedings* 2914(1), 030003. DOI: 10.1063/5.0175961